

IJSDS 2.0 Technical Architecture & Data Engineering Specification

Version 2.4.0-PROD

Updated Product Requirements / System Architecture Handoff

Revision	Change
2.4.0-PROD	Adds plagiarism detection, AI-authorship risk detection, machine-learning citation validation, citation graph recommendations, author-facing integrity reports, new data schema, audit orchestration, acceptance criteria, and cost-accounting requirements.
2.3.0-PROD	Baseline 28-day publication cycle, two-tier payment gate, reviewer referral engine, COPE conflict firewall, vector reviewer matching, and batch release engine.

Contents

1. Executive Summary & Core Engineering Principles
2. 28-Day Cycle Timeline & Pipeline Flow
3. Data Engineering & Infrastructure Architecture
4. Operational Delay, Integrity Audit & Citation Intelligence Specifications
5. Reviewer Matching & COPE Firewall SQL
6. Unit Economics Breakdown
7. Acceptance Criteria, Governance & Operational Requirements

IJSDS 2.0 Technical Architecture & Data Engineering Specification

System Name: International Journal of Social Work and Development Studies (IJSDS 2.0)

Document Type: Product Requirements / System Architecture & Data Pipeline Handoff Specification

Version: 2.4.0-PROD

Target Audience: CTO / Lead Data Engineer / Software Architect / Product & Editorial Operations

1. Executive Summary & Core Engineering Principles

IJSDS 2.0 is an automated, event-driven academic micro-publishing platform operating on a **28-Day Monthly Batch Publication Cycle**. It replaces legacy journal operations with an asynchronous multi-agent pipeline powered by a Model Cascade Architecture, a Two-Tier Split Payment Gate, an Ethically Safe Reviewer Referral Program, a research-integrity audit stack, citation intelligence, and a 28th-Day Monthly Release Engine.

Key System Targets & Operational Rules

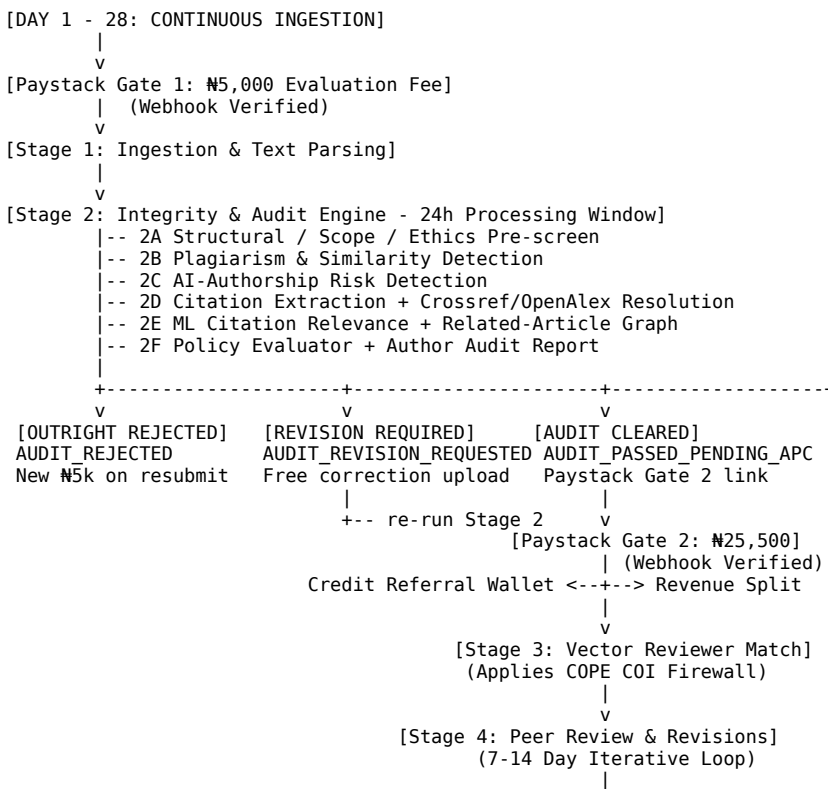
- **Continuous Submissions, Monthly Batch Releases:** Submissions are open 24/7/365. Publication occurs strictly on the 28th of every month.
 - **The 28th Batch Cutoff:** Manuscripts that pass all stages (APC paid, peer-reviewed, edited, formatted, DOI minted) by 23:59 WAT on the 27th are published in that month's issue on the 28th.

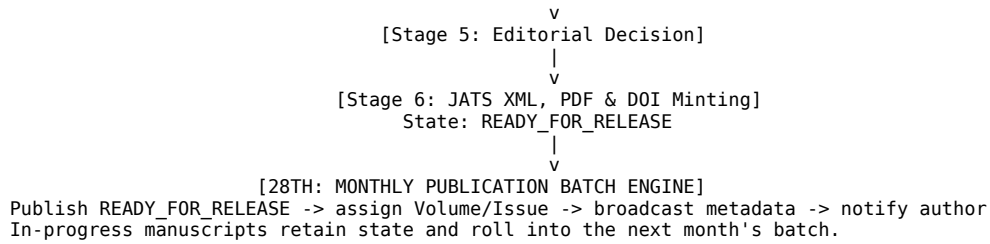
- **Automatic Rollover:** Manuscripts in progress (e.g., undergoing peer review or author corrections) roll over to the next month's batch without losing state.
- **24-Hour Audit Delay Algorithm:** The pre-screening and integrity audit uses a deterministic 24-hour processing window to smooth API load, permit Crossref/OpenAlex resolution, and provide a predictable SLA.
- **Two-Tier Financial Model:** Total Article Processing Charge (APC) is ₦30,500: Gate 1 is a ₦5,000 non-refundable Evaluation Fee; Gate 2 is a ₦25,500 APC collected only after the Stage 2 audit is cleared.
- **Fee Routing:** Gate 1 is routed in full to the Parakletus account. Gate 2 is split at settlement into ₦3,000 Tech Support, ₦1,000 Designer, ₦12,000 Journal Account, and ₦9,500 Journal Manager.
- **Reviewer Referral Incentive Engine:** Peer reviewers earn ₦1,000 per referred paper upon successful Gate 2 APC settlement. The referring reviewer is excluded from reviewing that manuscript.
- **Research Integrity Stack (NEW):** Stage 2 must run plagiarism/similarity detection, AI-authorship risk detection, citation extraction and validation, and a machine-learning reference relevance pass before an audit outcome is issued.
- **AI Detection Governance (NEW):** AI-detection output is a risk signal, not conclusive authorship evidence. It must not be the sole basis for automatic rejection. High-risk results trigger human/editorial review or a revision request according to configurable policy.
- **Citation Intelligence (NEW):** Every parsed reference must be resolved where possible against Crossref/OpenAlex, checked for metadata consistency, linked to its in-text citation context, and scored for semantic relevance. The system must also produce an author-facing list of related/cross-referenced articles.

Audit Outcome Rules (Gate 1 Handling)

- **Outright Rejection:** Severe confirmed plagiarism, core structural failure, irrelevance, or a major integrity failure confirmed by policy/editorial review. State -> AUDIT_REJECTED. Resubmission requires a new draft and a new ₦5,000 Evaluation Fee.
- **Revision Required:** Minor formatting issues, missing ethics statements, borderline similarity, unresolved/incorrect references, weak citation support, or AI-risk signals requiring author clarification. State -> AUDIT_REVISION_REQUESTED. Corrected manuscripts can be uploaded without a second ₦5,000 fee.
- **Audit Cleared:** All mandatory integrity checks are complete and no blocking issue remains. State -> AUDIT_PASSED_PENDING_APC and a Paystack link for ₦25,500 is generated.

2. 28-Day Cycle Timeline & Pipeline Flow





Stage 2 Required Outputs

Output	Purpose	Author-visible?
Plagiarism / similarity evidence	Matched sources, passage-level overlap, exclusions for bibliography/quotes where supported, overall risk classification.	Yes - summarized with evidence links/identifiers.
AI-authorship risk signal	Provider/model/version, document and segment scores, confidence/risk band, and any human-review flag.	Yes - preferably risk band and flagged passages; raw score display is configurable.
Citation validation ledger	Each in-text citation mapped to a bibliography record and, where possible, a resolved Crossref/OpenAlex work/DOI.	Yes.
Citation relevance score	Semantic relationship between citation context and cited work metadata/abstract; used to prioritize review, not to auto-rewrite citations.	Yes - as warnings/recommendations.
Related/cross-referenced article list	Ranked articles connected by reference graph and semantic similarity, including title, year, DOI/URL identifier, and why it is relevant.	Yes - delivered with the audit report.
Audit decision payload	Machine signals, policy rule hits, human-review requirement, final audit outcome, and complete versioned provenance.	No raw internals; author receives decision and actionable issues.

3. Data Engineering & Infrastructure Architecture

DDL Schema Updates (PostgreSQL 16+)

```

-- Enable Extensions
CREATE EXTENSION IF NOT EXISTS "uuid-oss";
CREATE EXTENSION IF NOT EXISTS "vector";

-- Existing manuscript state machine remains unchanged
CREATE TYPE manuscript_state AS ENUM (
  'DRAFT_PENDING_FEE1', 'SUBMITTED', 'IN_AUDIT', 'AUDIT_REJECTED',
  'AUDIT_REVISION_REQUESTED', 'AUDIT_PASSED_PENDING_APC', 'APC_PAID',
  'REVIEWER_MATCHED', 'IN_PEER_REVIEW', 'AUTHOR_REVISION_REQUESTED',
  'ACCEPTED', 'IN_PRODUCTION', 'READY_FOR_RELEASE', 'PUBLISHED'
);

```

Existing Core Tables (retained)

```

CREATE TABLE reviewer_profiles (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  user_id UUID NOT NULL UNIQUE,
  referral_code VARCHAR(20) UNIQUE NOT NULL,
  wallet_balance_naira NUMERIC(10, 2) DEFAULT 0.00,
  expertise_keywords TEXT[],
  expertise_vector vector(1536),
  is_available BOOLEAN DEFAULT TRUE,
  active_load INT DEFAULT 0
);

```

```

CREATE INDEX idx_reviewer_vector ON reviewer_profiles
USING hns (expertise_vector vector_cosine_ops);

CREATE TABLE manuscripts (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  title TEXT NOT NULL,
  abstract TEXT NOT NULL,
  author_id UUID NOT NULL,
  referred_by_reviewer_id UUID REFERENCES reviewer_profiles(id),
  current_state manuscript_state NOT NULL DEFAULT 'DRAFT_PENDING_FEE1',
  raw_file_r2_key TEXT NOT NULL,
  parsed_payload JSONB DEFAULT '{}::jsonb',
  abstract_vector vector(1536),
  doi TEXT UNIQUE,
  audit_scheduled_for TIMESTAMPTZ,
  audit_completed_at TIMESTAMPTZ,
  target_publication_year INT,
  target_publication_month INT,
  published_at TIMESTAMPTZ,
  audit_revision_count INT DEFAULT 0,
  created_at TIMESTAMPTZ DEFAULT NOW(),
  updated_at TIMESTAMPTZ DEFAULT NOW()
);

```

NEW - Integrity Audit Run Ledger

```

CREATE TYPE audit_run_status AS ENUM ('QUEUED', 'RUNNING', 'COMPLETED', 'FAILED', 'MANUAL_REVIEW');

CREATE TABLE manuscript_integrity_audits (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  manuscript_id UUID NOT NULL REFERENCES manuscripts(id) ON DELETE CASCADE,
  revision_no INT NOT NULL,
  status audit_run_status NOT NULL DEFAULT 'QUEUED',
  plagiarism_score NUMERIC(6,5),
  plagiarism_risk VARCHAR(20),
  plagiarism_provider VARCHAR(80),
  ai_risk_score NUMERIC(6,5),
  ai_risk_band VARCHAR(20),
  ai_detector_provider VARCHAR(80),
  ai_detector_model_version VARCHAR(120),
  citation_validation_score NUMERIC(6,5),
  unresolved_citation_count INT DEFAULT 0,
  citation_warning_count INT DEFAULT 0,
  requires_human_review BOOLEAN DEFAULT FALSE,
  policy_rules_hit JSONB DEFAULT '[]::jsonb',
  evidence_payload JSONB DEFAULT '{}::jsonb',
  author_report_r2_key TEXT,
  started_at TIMESTAMPTZ,
  completed_at TIMESTAMPTZ,
  created_at TIMESTAMPTZ DEFAULT NOW(),
  UNIQUE(manuscript_id, revision_no)
);

```

NEW - Citation Resolution & Validation Ledger

```

CREATE TYPE citation_resolution_status AS ENUM (
  'VALIDATED', 'PARTIAL_MATCH', 'UNRESOLVED', 'METADATA_MISMATCH'
);

CREATE TABLE manuscript_citations (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  manuscript_id UUID NOT NULL REFERENCES manuscripts(id) ON DELETE CASCADE,
  audit_id UUID NOT NULL REFERENCES manuscript_integrity_audits(id) ON DELETE CASCADE,
  bibliography_index INT,
  raw_reference TEXT NOT NULL,
  in_text_contexts JSONB DEFAULT '[]::jsonb',
  normalized_title TEXT,
  normalized_authors JSONB DEFAULT '[]::jsonb',
  publication_year INT,
  supplied_doi TEXT,
  resolved_doi TEXT,
  crossref_work_id TEXT,
  openalex_work_id TEXT,
  resolution_status citation_resolution_status NOT NULL,
  metadata_match_score NUMERIC(6,5),
  semantic_relevance_score NUMERIC(6,5),
  citation_context_vector vector(1536),
  warning_codes TEXT[],
  created_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE INDEX idx_citation_context_vector ON manuscript_citations
USING hns (citation_context_vector vector_cosine_ops);

```

NEW - Related / Cross-Referenced Article Recommendations

```
CREATE TABLE citation_recommendations (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  manuscript_id UUID NOT NULL REFERENCES manuscripts(id) ON DELETE CASCADE,
  audit_id UUID NOT NULL REFERENCES manuscript_integrity_audits(id) ON DELETE CASCADE,
  source VARCHAR(30) NOT NULL, -- CROSSREF | OPENALEX | LOCAL_INDEX
  external_work_id TEXT,
  doi TEXT,
  title TEXT NOT NULL,
  publication_year INT,
  relevance_score NUMERIC(6,5) NOT NULL,
  graph_proximity_score NUMERIC(6,5),
  recommendation_reason TEXT,
  rank_position INT NOT NULL,
  selected_by_author BOOLEAN,
  created_at TIMESTAMPTZ DEFAULT NOW()
);
```

Existing Payment / Referral Tables (retained)

```
CREATE TABLE referral_payout_ledger (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  reviewer_id UUID REFERENCES reviewer_profiles(id) ON DELETE CASCADE,
  manuscript_id UUID REFERENCES manuscripts(id) ON DELETE CASCADE,
  amount_naira NUMERIC(10, 2) DEFAULT 1000.00,
  status VARCHAR(20) CHECK (status IN ('PENDING', 'CREDITED', 'WITHDRAWN')),
  created_at TIMESTAMPTZ DEFAULT NOW()
);
```

```
CREATE TABLE payment_ledger (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  manuscript_id UUID REFERENCES manuscripts(id) ON DELETE CASCADE,
  gateway_reference TEXT UNIQUE NOT NULL,
  payment_type VARCHAR(20) CHECK (payment_type IN ('EVALUATION_FEE', 'APC')),
  amount_naira NUMERIC(10, 2) NOT NULL,
  status VARCHAR(20) NOT NULL,
  raw_webhook_payload JSONB,
  created_at TIMESTAMPTZ DEFAULT NOW()
);
```

```
CREATE TABLE gate2_distribution_ledger (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  manuscript_id UUID REFERENCES manuscripts(id) ON DELETE CASCADE,
  payment_ledger_id UUID REFERENCES payment_ledger(id) ON DELETE CASCADE,
  tech_support_naira NUMERIC(10, 2) NOT NULL DEFAULT 3000.00,
  designer_naira NUMERIC(10, 2) NOT NULL DEFAULT 1000.00,
  journal_account_naira NUMERIC(10, 2) NOT NULL DEFAULT 12000.00,
  journal_manager_naira NUMERIC(10, 2) NOT NULL DEFAULT 9500.00,
  created_at TIMESTAMPTZ DEFAULT NOW(),
  CONSTRAINT chk_gate2_split_totals_25500 CHECK (
    tech_support_naira + designer_naira + journal_account_naira + journal_manager_naira = 25500.00
  )
);
```

4. Operational Delay, Integrity Audit & Citation Intelligence Specifications

A. The 24-Hour Synthetic Audit Delay Algorithm

To ensure system stability, accommodate rate limits for Crossref/OpenAlex and external integrity providers, and set clear expectations for authors, the audit engine executes on a 24-hour delayed queue pattern. The delay is an SLA window, not an instruction to idle: background parsing, caching, reference resolution, and provider retries may run during the window.

```
from datetime import datetime, timedelta

def queue_integrity_audit(manuscript_id: str):
    scheduled_time = datetime.utcnow() + timedelta(hours=24)
    db.execute(
        "UPDATE manuscripts SET audit_scheduled_for=%s, current_state='IN_AUDIT' WHERE id=%s",
        (scheduled_time, manuscript_id)
    )
    audit_worker.apply_async(args=[manuscript_id], eta=scheduled_time)
```

B. Stage 2 Integrity Orchestrator (NEW)

```
def execute_integrity_audit(manuscript_id):
    manuscript = load_parsed_manuscript(manuscript_id)
    audit = create_versioned_audit_run(manuscript)
```

```

structural = run_structural_scope_ethics_checks(manuscript)
plagiarism = plagiarism_adapter.check(manuscript.full_text)
ai_signal = ai_detection_adapter.score(manuscript.full_text)

refs = extract_references_and_in_text_contexts(manuscript)
resolved = resolve_references(refs, providers=[crossref, openalex])
citation_scores = score_citation_relevance(resolved, manuscript)
related = recommend_related_articles(
    manuscript=manuscript,
    resolved_citations=resolved,
    graph_sources=[crossref, openalex, local_index]
)

decision = policy_engine.evaluate(
    structural=structural,
    plagiarism=plagiarism,
    ai_signal=ai_signal,
    citations=citation_scores
)

# AI detection must never be a sole auto-rejection rule.
if decision.reason == 'AI_SIGNAL_ONLY':
    decision = 'MANUAL_REVIEW_OR_REVISION'

report = build_author_integrity_report(
    plagiarism, ai_signal, resolved, citation_scores, related, decision
)
persist_audit_evidence(audit, report, decision)
send_author_audit_report(manuscript.author_id, report)
transition_manuscript_state(manuscript_id, decision)

```

C. Machine-Learning Referencing Algorithm (NEW)

The referencing engine is a hybrid deterministic + machine-learning pipeline. Deterministic metadata resolution establishes identity; ML ranks relevance and detects weak or suspicious citation relationships. The engine must not silently add, remove, or rewrite citations on the author's behalf.

1. Citation extraction: detect bibliography entries, citation markers, footnote citations, DOI/URL identifiers, and the sentence/paragraph context in which each citation is used.
2. Metadata normalization: canonicalize title, author names, year, journal, volume/issue, DOI and URL fields.
3. Identity resolution: query Crossref/OpenAlex and the local cache. Choose the best candidate using DOI exact match first, then normalized metadata similarity.
4. Context embedding: create an embedding for each in-text citation context and, where available, for the cited work title/abstract/keywords.
5. Relevance scoring: combine semantic similarity, metadata confidence, graph proximity, and duplicate/self-citation signals into a review score. Thresholds are configuration, not hard-coded constants.
6. Cross-reference discovery: expand from resolved references through cited-by/reference/related-work graph links and rank candidate articles against the manuscript abstract/sections.
7. Author output: deliver validated references, unresolved/mismatched references, weak-support warnings, and a ranked list of related/cross-referenced articles with stable identifiers.
8. Feedback loop: log author/editor dispositions (accepted warning, corrected citation, dismissed warning, selected related paper) for future supervised calibration without changing historical audit results.

D. Author-Facing Integrity & Citation Report (NEW)

Section	Required Contents
Audit decision	Cleared / Revision Required / Rejected, plus actionable reason codes.
Plagiarism	Overall risk, matched sources, overlap snippets or locations, and exclusions applied.
AI-authorship risk	Risk band, detector/provider/version, flagged sections, and a clear statement that the signal is not conclusive evidence by itself.
Citation validation	Validated references, DOI/metadata matches, unresolved references, metadata mismatches, duplicate references and missing in-text/bibliography mappings.

Citation relevance	Warnings where the cited work appears weakly related to the surrounding claim/context; editorial review remains authoritative.
Related literature	Ranked related/cross-referenced article list from Crossref/OpenAlex/local index, with identifiers and recommendation rationale.
Required actions	Specific corrections needed before resubmission. Revision upload re-triggers the 24-hour Stage 2 audit without another Gate 1 fee.

E. 28th Monthly Batch Release Engine

```
def execute_monthly_publication_batch():
    eligible_manuscripts = db.query(
        "SELECT id FROM manuscripts WHERE current_state='READY_FOR_RELEASE'"
    )
    for paper in eligible_manuscripts:
        with db.transaction():
            mark_published_and_assign_issue(paper.id)
            broadcast_to_crossref_and_indexers(paper.id)
            send_publication_notice_email(paper.id)
# Papers still in review/revision remain untouched and roll over naturally.
```

5. Reviewer Matching & COPE Firewall SQL

When matching reviewers at Stage 3, vector search strictly excludes the referring reviewer. The existing COPE conflict firewall remains unchanged by the v2.4 integrity additions.

```
SELECT
    rp.id AS reviewer_id,
    1 - (rp.expertise_vector <=> m.abstract_vector) AS similarity_score
FROM reviewer_profiles rp
JOIN manuscripts m ON m.id = :target_manuscript_id
WHERE rp.is_available = TRUE
    AND rp.active_load < 3
    AND rp.id != COALESCE(
        m.referred_by_reviewer_id,
        '00000000-0000-0000-0000-000000000000'::uuid
    )
ORDER BY rp.expertise_vector <=> m.abstract_vector
LIMIT 5;
```

6. Unit Economics Breakdown

Existing Revenue Model Per Paper

Item	Amount / Rule
Gate 1 Evaluation Fee	₦5,000 - non-refundable; routed to Parakletus.
Gate 2 APC	₦25,500 - collected only after audit clearance.
Reviewer referral commission	₦1,000 per qualifying referred paper upon Gate 2 settlement.
Gate 2 internal split	₦3,000 Tech Support; ₦1,000 Designer; ₦12,000 Journal Account; ₦9,500 Journal Manager.

v2.4 Cost Treatment for New Integrity Services

The previous v2.3 estimate treated the integrity audit as a very low fixed cost. That assumption is no longer safe once commercial plagiarism/AI-detection services and expanded reference-resolution calls are introduced. v2.4 therefore makes integrity cost provider-dependent and metered rather than hard-coding a new estimate before vendor selection and load testing.

```

legacy_full_workflow_cogs_naira = 1259.00
legacy_integrity_line_naira = 15.00
base_cogs_excluding_integrity = 1244.00

```

```

updated_cogs = base_cogs_excluding_integrity + integrity_stack_cost_naira
updated_net_before_internal_revenue_accounting = 30500.00 - updated_cogs
updated_gross_margin_pct = updated_net_before_internal_revenue_accounting / 30500.00 * 100

```

- **Meter each provider call per audit run:** plagiarism, AI detection, embeddings, Crossref/OpenAlex lookups, and report-generation model usage.
- **Cache DOI/metadata resolution results** to avoid repeated external lookup charges and reduce latency.
- **Do not publish a revised margin percentage** in production documentation until the chosen providers have been benchmarked against real manuscript lengths and revision frequency.

7. Acceptance Criteria, Governance & Operational Requirements

Functional Acceptance Criteria

Capability	Acceptance condition
Plagiarism detection	Every Stage 2 run stores a versioned similarity result and source evidence. A revision re-runs the check and preserves prior results.
AI detection	Every Stage 2 run stores provider/model/version and segment-level or document-level risk output. No rule may auto-reject solely because of the AI signal.
Citation mapping	The system links in-text citation markers to bibliography entries where structurally possible and flags orphaned markers/uncited bibliography entries.
Reference resolution	Each reference gets a resolution status and stable external identifier where available. Unresolved or mismatched references appear in the author report.
ML relevance	Citation-context relevance is scored and stored, with explainable warning codes. The score does not silently alter author references.
Related article list	Each completed audit can generate a ranked list of relevant Crossref/OpenAlex/local-index articles and include that list in the author-facing report.
Revision loop	Revision uploads do not charge Gate 1 again and re-trigger a new versioned audit run under the 24-hour Stage 2 SLA.
Audit reproducibility	Historical audits retain provider/model/version, policy rules, timestamps and evidence so the decision can be reconstructed later.
Failure handling	Provider timeouts/retries must not silently clear a manuscript. Failed mandatory checks keep the audit incomplete or route it to manual review.
Release protection	No manuscript can transition to AUDIT_PASSED_PENDING_APC unless all mandatory Stage 2 components are completed or an authorized manual override is recorded.

Governance & Safety Requirements

- **Provider-agnostic adapters:** plagiarism and AI detection must sit behind internal interfaces so vendors can be replaced without changing workflow/state logic.
- **Configurable thresholds:** similarity, AI-risk and citation-relevance thresholds must be configuration data with change history, not constants buried in code.
- **Human-review override:** editorial staff can resolve disputed flags, but the override must be attributed and auditable.

- **Data minimization:** where an external service is used, send only the content needed for the check and retain provider request IDs rather than unnecessary duplicate manuscript copies.
- **No fabricated references:** recommendation output must originate from resolved external/local records and must include stable identifiers. The LLM layer may summarize relevance but may not invent bibliographic metadata.
- **Author control:** recommended related papers are suggestions only. The system does not insert them into the manuscript or bibliography without author action.

Implementation Sequence

1. Build provider-agnostic plagiarism and AI-detection adapters and the manuscript_integrity_audits ledger.
2. Implement reference extraction, Crossref/OpenAlex normalization/resolution and citation ledger persistence.
3. Add embeddings, citation-context relevance scoring and graph-based related-paper ranking.
4. Add policy evaluator rules and human-review routing, ensuring AI signal alone cannot reject.
5. Generate the author-facing audit/citation report and wire it to email/portal delivery.
6. Instrument per-provider cost, latency, error rate and cache hit rate; benchmark before finalizing unit-economics targets.
7. Run end-to-end acceptance tests across first submission, revision loop, audit rejection, audit clearance, Gate 2 payment and monthly release.

Configuration Decisions to Lock Before Production

Decision	Recommended PRD stance
Plagiarism provider	Keep vendor-agnostic in the PRD; select after accuracy, coverage, privacy, API and cost benchmark.
AI detection provider/model	Keep vendor-agnostic and version every result. Treat as review signal only.
Score thresholds	Store in versioned configuration; determine via editorial calibration rather than hard-coding in this document.
Related-paper list size	Make configurable by journal/editorial requirement; do not hard-code into the ranking algorithm.
Author visibility of raw AI score	Product/editorial setting. Minimum requirement is risk band + flagged sections + explanation of limitations.
Manual override role	Restrict to authorized editorial/technical roles and require reason logging.

Version 2.4.0-PROD change note: This revision preserves the v2.3 payment, reviewer, state-machine and 28th-day release design while expanding Stage 2 into a full research-integrity and citation-intelligence layer.